School of Mathematical Sciences | Lancaster University

Micro-macro changepoint inference for periodic data sequences

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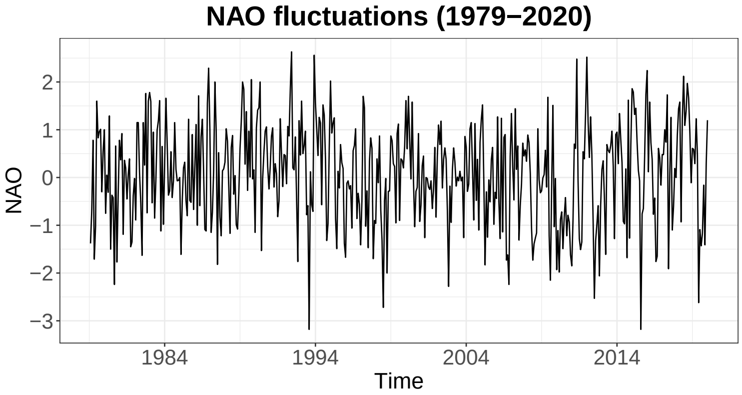
Outline

- Motivation
- What are changepoints?
- Micro changepoint model
    - Bayesian Implementation
    - Frequentist Implementation
- Macro changepoint model
- Simulations
- Application

Motivation

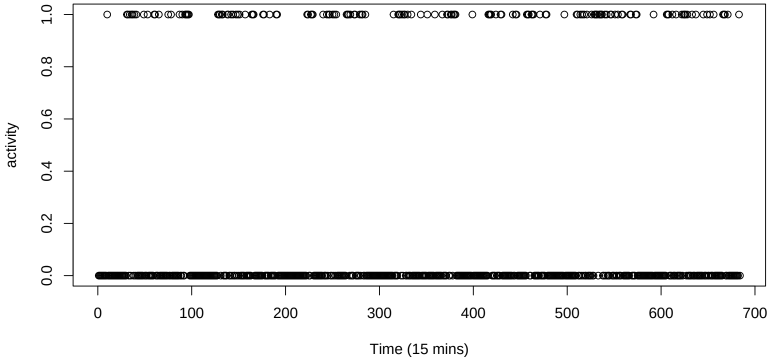
NAO

- How can we make the most of highly granular time series?
- How can we study changes in periodic and cyclical behaviours?



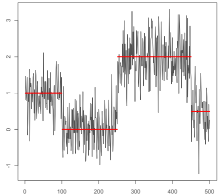
Howz

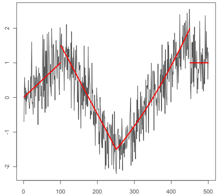
- How can we study changes in periodic and cyclical behaviours?

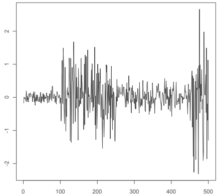


What are changepoints?

For data $y_1, \ldots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \ldots, y_\tau$ are different from $y_{\tau+1}, \ldots, y_n$ in some way. Traditional changepoints include: mean, regression, variance.





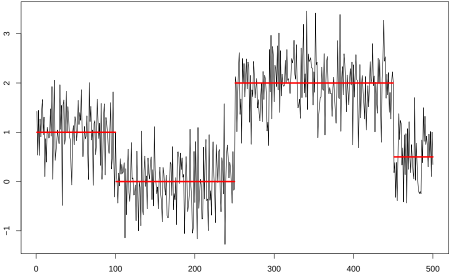


Change in mean

Assume we have time-series data where

$$Y_t|\theta_t \sim \mathrm{N}(\theta_t, 1),$$

but where the means, θ_t, are piecewise constant through time.



Inferring Changepoints

We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau}\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β. If we infer a changepoint its position is estimated as

$$\tau = \arg\max\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

This can test can be repeatedly applied to new segments to find multiple changepoints.

Likelihood Ratio Tests

Define m to be the number of changepoints, with positions $\boldsymbol{\tau} = (\tau_0, \tau_1, \ldots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

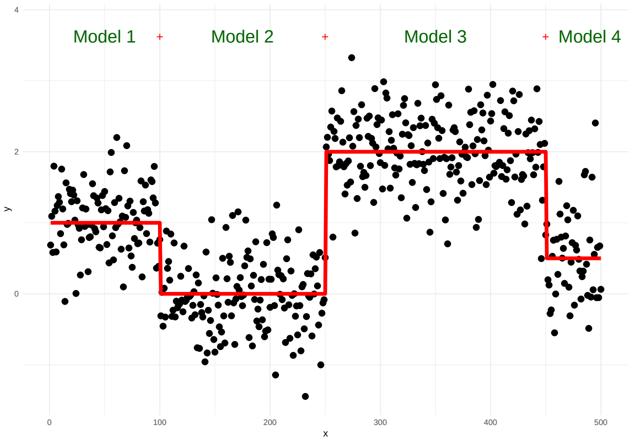
Then one application of the Likelihood ratio test can be viewed as

$$\min_{m \in \{0,1\}, \boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m, \boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

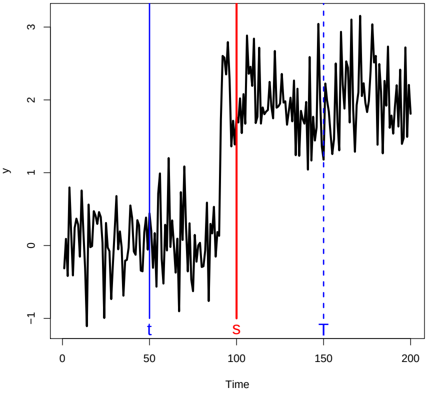
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell

- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.

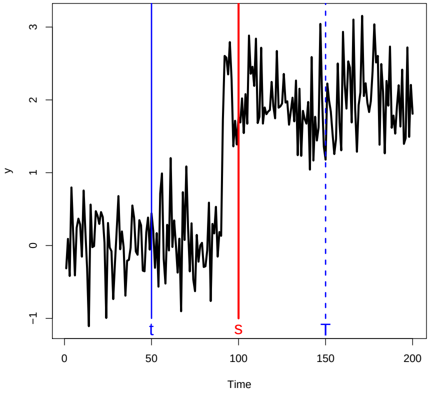


PELT: Pruning

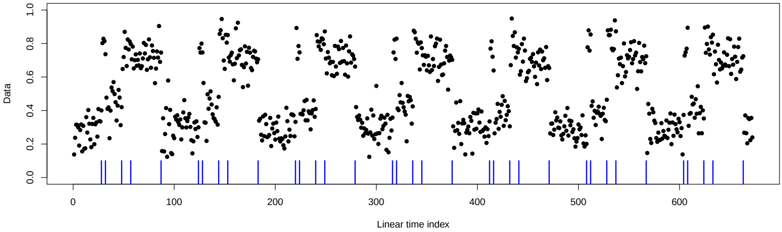
Let $0 < t < s < T$, if

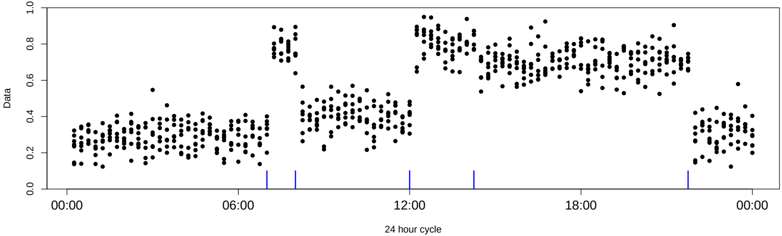
$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t can never be the optimal last changepoint prior to T.
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.



Periodic Changepoints





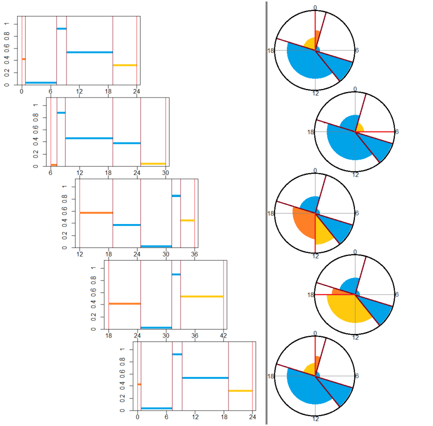
Existing Methods

- Binary segmentation & PELT
  - Assumes conditional independence ...
  - ... but dependence exists across days.

- Multivariate time series
  - Stack days to align changepoints ...
  - ... but what should ‘time zero’ be?

Existing Methods

- Circular changepoint
  - For circular data (e.g. direction) ...
  - ... but time axis is linear.
- Circular Binary Segmentation
  - local time axis is circular ...
  - ... but just joins up ends not each day.
- Cyclic-HMM – Achieves periodicity via a structured transition matrix ... but period duration is stochastic.



Our Contribution

Create method for periodic and global changepoint detection

- Wide range of data structures (assume likelihood available)
- Present Bayesian (RJMCMC fit) and frequentist for periodic
- Use PELT to extend to periodic

Create layered visualizations for ease of inference.

We denote changepoints as micro (periodic) and macro (global).

Periodic level changepoint detection

Thinking

Challenge: How to encorporate circular nature

Soution: Reframe $t = cN + i$, $c \in \mathbb{N}_0$ and $i \in \{0, 1, \ldots, N-1\}$

Challenge: How to detect using (c, i) instead of t?

Challenge: There is no single changepoint setting anymore!

Model

- $\{X_t\}_{t\geq 0}$ — Real-valued time series.
- $N \in \mathbb{N}$ — Given period length.
  - $t = cN + i$ where $c \in \mathbb{N}_0$ and $i \in \{0, 1, \ldots, N-1\}$.
- $m \geq 1$ — Number of segments partitioning the period.
- $\tau_1, \ldots, \tau_m \in \{0, 1, \ldots, N-1\}$ — Changepoint positions.
  - Ordered: $\tau_{j-1} < \tau_j$.
  - Periodic: $\tau_{m+1} = \tau_1$.
  - Minimum separation: $\tau_j - \tau_{j-1} \geq l$ for $l > 0$.
- μ_j for $j \in \{1, \ldots, m\}$ — Segment means.
- σ_j^2 for $j \in \{1, \ldots, m\}$ — Segment variances.

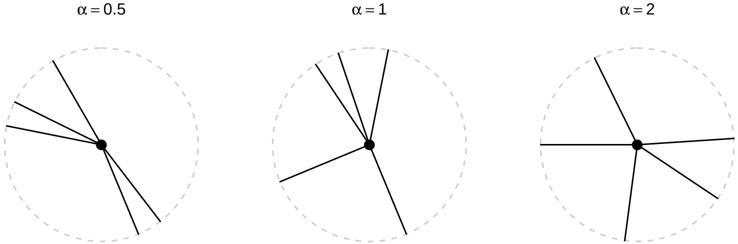
$X_t \sim \mathrm{Normal}\left(\mu(t), \sigma^2(t)\right)$ for $t \geq 0$ where
$\mu(t) = \sum_{j=1}^m \mu_j \mathbb{I}\left\{i \in (\tau_{j-1}, \tau_j]\right\}$ and $\sigma^2(t) = \sum_{j=1}^m \sigma_j^2 \mathbb{I}\left\{i \in (\tau_{j-1}, \tau_j]\right\}$.

Bayesian

Prior

- $m \sim \mathrm{Truncated-Poisson}\left(1,\ \{1, \ldots, \lfloor \frac{N}{l} \rfloor\}\right)$.
- $\mu_j | \sigma_j^2, \boldsymbol{\tau}, m \sim \text{Normal}(d, c\sigma_j^2)$ for $j = 1, \ldots, m$.
- $\sigma_j^2 | \boldsymbol{\tau}, m \sim \text{InvGamma}(a, b)$ for $j = 1, \ldots, m$.
- Transform $\boldsymbol{\tau}$ to excess segment length:
  - $\delta_j = \tau_j - \tau_{j-1} - l$ for $j = 1, \ldots, m$.
  - $(\delta_1, \ldots, \delta_m) \sim \mathrm{Dirichlet-Multinomial}(N - lm,\ \alpha \mathbf{1}_m)$
- $\tau_1 \sim \mathrm{Uniform}\{0, \ldots, N-1\}$.

Cluster | Random | Repel



Posterior

Segment probabilities: Posteriors conditional on a sampled number and location of changepoints are easy to compute via conjugacy.

Number of segments and changepoint positions:

$$\pi(\boldsymbol{\tau}, m|\boldsymbol{x}) \propto \begin{cases} B(1+\sum_t x_t,\ 1+\sum_t(1-x_t)) & m=1, \\ \prod_{j=1}^m \left[\frac{B(1+s_j,\ 1+n_j-s_j)}{(\alpha+\delta_j)B(\alpha,\ \delta_j+1)}\right] \frac{(m\alpha+N-lm)B(m\alpha,\ N-lm+1)}{m!N} & \text{o/w}, \end{cases}$$

where $B(\cdot,\cdot)$ is the beta function.

- Trivial to evaluate $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$ for a given changepoint vector.
- Support size is potentially impractically large:
  - For $N = 96$ and $l = 4$, then there are $\sim 27.3 \times 10^{12}$ events.

RJMCMC

- Permits exploration of varying dimensional spaces.
- Need to satisfy ‘dimension-matching’ condition.
- Focus inference on the marginal posterior $\pi(\boldsymbol{\tau}, m|\boldsymbol{x})$.

Algorithm

1. Sample a segment $j \in \{1, \ldots, m\}$ uniformally.
2. Define the proposal set $\mathcal{P}_j(\boldsymbol{\tau}, m)$.
3. Sample a changepoint vector from the proposal pmf:

$$q(\boldsymbol{\tau}', m'|\boldsymbol{\tau}, m) \propto \pi(\boldsymbol{\tau}', m'|\boldsymbol{x})\mathbb{I}\left\{(\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m)\right\}.$$

4. Accept/Reject proposal. Return to 1.

- Evaluate $\pi(\phi|\boldsymbol{\tau}, m, \boldsymbol{x})$ given the changepoint posterior samples.

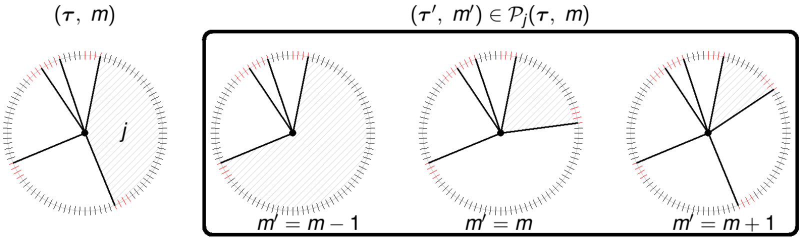
Proposal Set

Proposal set, $\mathcal{P}_j(\boldsymbol{\tau}, m)$, contains three move types:

- Combine sampled segment with the next.
- Contract or extend the length of the current segment.
- Partition current segment into two.

Unique mapping

For each $(\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m)$ excluding $(\boldsymbol{\tau}, m)$, there exists a unique segment j' such that $(\boldsymbol{\tau}, m) \in \mathcal{P}_{j'}(\boldsymbol{\tau}', m')$.



Frequentist

Segment Neighbourhood

Challenge: No single changepoint setting to build upon.
Standard:

$$c_{0,t}^{m} = \min_{\tau_1,\dots,\tau_m} \sum_{j=1}^{m+1} \mathcal{C}(x_{(\tau_{j-1}+1):\tau_j})$$
$$= \min_{\tau_{m-1}} [c_{0,\tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1):t})].$$

Circular:

$$c_{0,t}^{2} = \min_{\tau_1,\tau_2} [\mathcal{C}(x_{(\tau_1+1):\tau_2}) + \mathcal{C}(x_{(\tau_2+1):\tau_1})]$$
$$c_{0,t}^{m} = \min_{\tau_{m-1}} [c_{0,\tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1):t})] \qquad m = 3, \dots M$$

Bayesian

Correct detection (1-α and power rates)

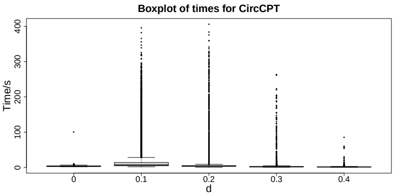
d \ τ_1	4	8	16	24	32	48
0.0	1.000	0.998	1.000	1.000	0.999	0.999
0.1	**0.003**	**0.009**	**0.073**	**0.130**	**0.134**	**0.163**
0.2	**0.365**	**0.516**	**0.544**	0.535	0.514	0.552
0.3	**0.891**	0.841	0.786	0.806	0.826	0.827
0.4	0.970	0.951	0.897	0.917	0.912	0.915

Frequentist

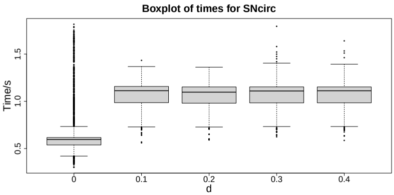
Correct detection (1-α and power rates)

d \ τ_1	4	8	16	24	32	48
0.0	1.000	**1.000**	1.000	1.000	**1.000**	**1.000**
0.1	0.000	0.004	0.042	0.102	0.123	0.156
0.2	0.177	0.467	0.550	**0.540**	**0.519**	**0.570**
0.3	0.842	**0.846**	**0.828**	**0.856**	**0.866**	**0.873**
0.4	**0.973**	**0.975**	**0.971**	**0.972**	**0.966**	**0.976**

Simulation Results

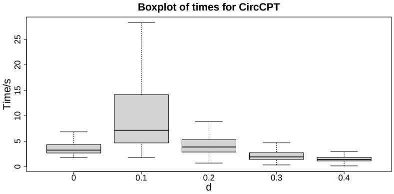


Bayesian

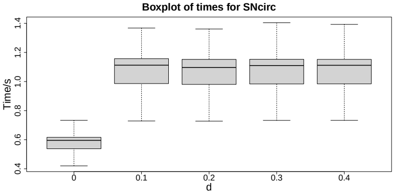


Frequentist

Simulation Results



Bayesian



Frequentist

Macro Level Changepoint Detection

Macro Level Changepoints

- Treat the micro-level detection as a "fit" for a segment ...
- ... thus we can use traditional, linear time algorithms.

We use PELT which optimizes:

$$\sum_{i=1}^{q+1}[\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i})] + \beta q. \tag{1}$$

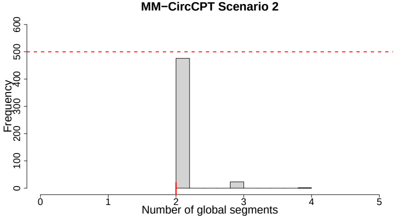
where q is the number of macro level changepoints and $\mathcal{C}(\cdot)$ is the (MAP) likelihood:

$$\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i}) = -2\sum_{j=1}^{\hat{m}_i}\sum_{t\in B^*_{i,j}} \log f(y_t|\hat{\mu}_{i,j}, \hat{\sigma}^2_{i,j})$$

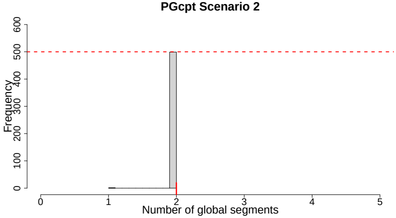
Simulations

See pre-print for full simulation study.

Simulation Results

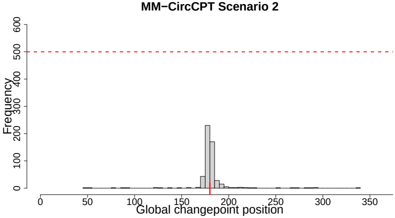


Bayesian

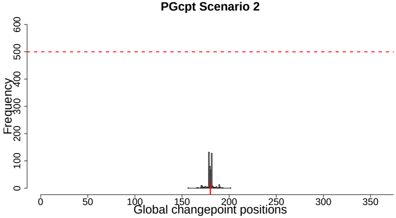


Frequentist

Simulation Results

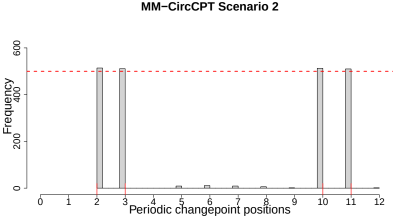


Bayesian

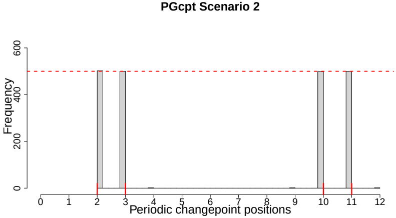


Frequentist

Simulation Results



Bayesian



Frequentist

Application

Application Results:NAO

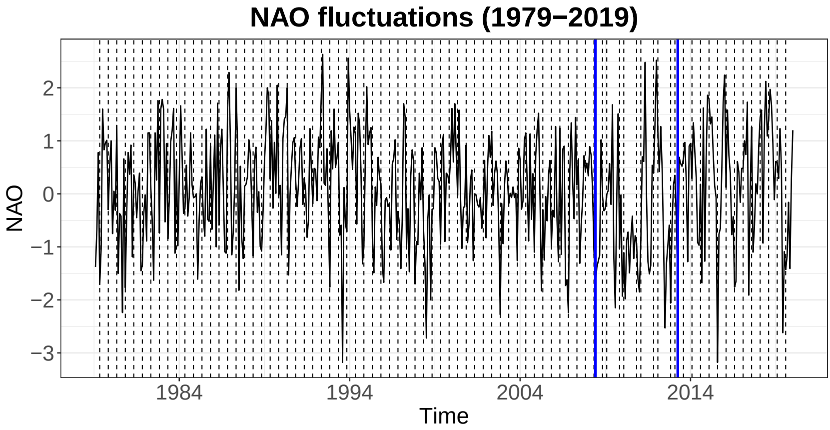


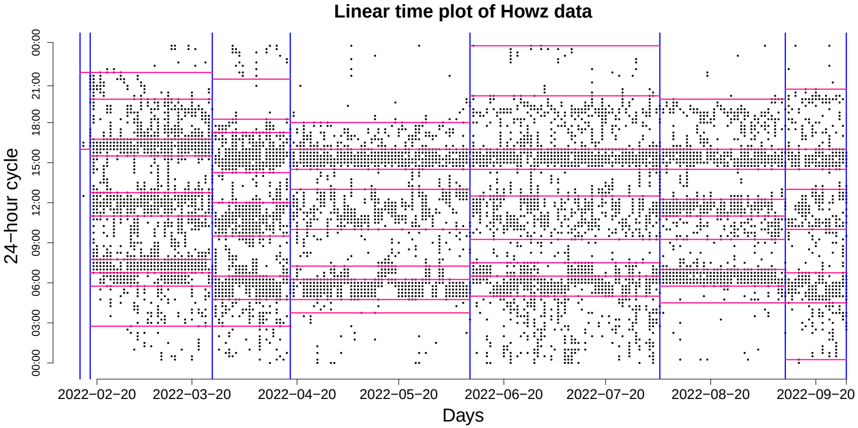
Figure: NAO data and Changepoint positions using $pen = 4\log(n)$

Application Results:NAO

Macro-level segment	Micro-level changepoints
1979-2008	April and October
2008-2013	May and August
2013-2019	April and October

Table: Macro- and micro-level changepoint positions for the NAO index.

Application Results:Howz



Discussion

Overview:

- A new approach for identifying within-period changepoint events for time series.

  Taylor, S. A. C., Killick, R., Burr, J. and Rogerson, L. (2021)

  'Changepoint Detection within Periodic Binary Time Series'. JRSS:Series C.
- Pooling of evidence across multiple periods by applying a circular perspective to time.
- Introduced the micro-macro level changepoint approach

  Ushakova, A., Taylor, S. A. C., Killick, R. (2023)

  'Micro-Macro Changepoint Inference for Periodic Data Sequences'. JCGS.
- Bayesian and Frequentist viewpoints for a wide set of data types

  Li, O., Killick, R. (2025+)

  'Detecting changes in periodic data'. Submitted

WBS performance

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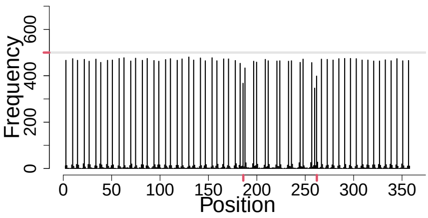
Scenario 11



Scenario 7